

# Zachary Benson

Senior full stack engineer (React, Angular, Node, AWS, TypeScript)

Portland, Oregon, US

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## PROFESSIONAL SUMMARY

Versatile and results-oriented Senior Full Stack Engineer with 8+ years of experience architecting and delivering high-performance, scalable web applications using React, Node.js, Vue.js, Angular, and TypeScript. Skilled in both frontend and backend development, with deep expertise in building responsive UIs, secure APIs, and modular, maintainable architectures. Proficient in integrating RESTful and GraphQL APIs, optimizing performance through lazy loading, code splitting, and SSR/SSG strategies. Experienced with cloud infrastructure using AWS, Docker, and Kubernetes, and implementing robust CI/CD pipelines for efficient deployment. Strong background in Agile/Scrum teams, leading full project lifecycles, mentoring developers, and enforcing best practices in code quality, testing, and documentation. Passionate about solving real-world problems, aligning technology with business goals, and delivering impactful, user-focused digital experiences across the stack.

## EDUCATION

University of Florida - Bachelor's Degree in computer Engineering, 2014 - June 2018

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## TECHNICAL SKILLS

**Frontend Frameworks:** Advanced expertise in React.js, Next.js, Remix, Angular, Vue.js, and Nuxt.js, building high-performance SPAs and SSR apps with accessibility and responsiveness in mind.

**State Management:** Proficient in Redux, Zustand, Context API, MobX, Vuex, and Recoil for building scalable, maintainable state architectures across complex UIs.

**Component Libraries & Design Systems:** Skilled in customizing and extending MUI, Chakra UI, Ant Design, and Bootstrap. Experienced in building design systems and UI component libraries for enterprise-scale reuse.

**Backend & Microservices Development:** Expertise in Node.js, NestJS, Express.js, and TypeScript for scalable backend architecture. Experience with event-driven, microservices, and domain-driven design (DDD) patterns.

**API Architecture:** Architected and consumed RESTful, GraphQL, and gRPC APIs; integrated Apollo Server/Client, Swagger/OpenAPI, Postman, and AsyncAPI for robust API documentation and testing.

**Cloud & DevOps:** Deep hands-on experience with AWS (Lambda, S3, RDS, DynamoDB, API Gateway, CloudFront, ECS) and Azure cloud services. Proficient in Terraform and CloudFormation for Infrastructure as Code (IaC). Built robust CI/CD pipelines using GitHub Actions, Jenkins, GitLab CI, and CircleCI. Dockerized services and deployed with Kubernetes, ECS, and Helm.

**Messaging & Event Streaming:** Experience with Apache Kafka, RabbitMQ, and Amazon SNS/SQS to enable asynchronous communication, decoupling, and real-time data streaming in distributed microservices systems.

**Testing & Quality Assurance:** Advanced testing with Jest, Cypress, React Testing Library, Supertest, and Mocha for TDD/BDD. Experience setting up test coverage pipelines and contract testing with Pact.

**Performance Optimization:** Skilled in lazy loading, code splitting, tree shaking, Webpack/Babel optimization, server-side caching (Redis, Varnish), and web vitals monitoring to ensure top-tier frontend and backend performance.

**Database Systems:** Hands-on with MongoDB, PostgreSQL, MySQL, and DynamoDB. Proficient with ORMs like Prisma, Sequelize, and TypeORM, implementing scalable, normalized, and performant data schemas.

**Security & Auth:** Implemented OAuth2, JWT, SAML, and OpenID Connect. Enforced RBAC, CORS, rate-limiting, CSRF/XSS protection, and secure API gateways.

**Agile Practices & Collaboration:** Strong experience in Agile/Scrum, SAFe, and Kanban environments. Collaborated via JIRA, Confluence, Slack, Miro, and Figma across globally distributed teams. Led architecture reviews and code reviews following best practices and design principles.

**Tools & IDEs:** Proficient in VS Code, WebStorm, Postman, Docker Desktop, K9s, Figma, and browser dev tools for full-cycle development, debugging, and testing.

## EXPERIENCE

### **Deloitte - Remote - Senior full stack engineer Feb 2022 – Dec 2025**

- Architected and delivered a high-performance, cloud-native e-commerce platform using the MERN stack (MongoDB, Express.js, React, Node.js) within a microservices architecture, ensuring scalability, modularity, and independent deployments.
- Led the design and architecture of RESTful and GraphQL APIs, establishing consistent patterns for versioning, error handling, authentication (JWT, OAuth2), and data pagination, enabling secure and scalable data exchange across services.
- Created a centralized API gateway pattern with rate limiting, request validation, and caching using Redis and API Gateway to improve latency and throughput across distributed services.
- Designed and built a highly scalable React component architecture using TypeScript, following atomic design principles and Storybook-driven development, supporting reusable, testable, and responsive UI components across multiple brands and devices.
- Implemented Next.js SSR/SSG strategies, dynamic imports, lazy loading, and code splitting to enhance frontend performance, Core Web Vitals, and SEO readiness.
- Defined and maintained backend modules with NestJS, applying domain-driven design, dependency injection, and Mongoose ORM, improving maintainability and separation of concerns.
- Integrated secure payment workflows with Stripe and PayPal, ensuring PCI-DSS compliance, and built extensible middleware for webhook handling and fraud detection.
- Built and deployed cloud infrastructure using AWS (EC2, S3, RDS, Lambda, CloudFront) and Terraform for Infrastructure as Code (IaC), supporting scalability, monitoring, and disaster recovery.
- Automated full CI/CD pipelines using GitHub Actions, Docker, Helm, and Kubernetes, enabling faster deployments with environment parity, canary releases, and rollback support.
- Implemented full-stack observability using Prometheus, ELK Stack, and AWS CloudWatch, including dashboards, alerts, and logging standards for production insights.
- Applied test-driven development (TDD) using Jest, Cypress, and React Testing Library, ensuring robust test coverage for unit, integration, and E2E flows.
- Collaborated with designers to translate Figma wireframes into responsive, accessible, and scalable UI systems, promoting consistent branding and usability across platforms.
- Authored technical documentation, API specifications (OpenAPI/Swagger), component libraries, and DevOps runbooks to support ongoing development and team knowledge sharing.
- Mentored frontend and backend engineers on scalable architecture, code quality, system performance, and agile delivery practices, contributing to a high-performing, cross-functional team.

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## **General Motors – Florida, Senior full stack engineer Oct 2019 - Feb 2022**

- Architected and developed a real-time vehicle tracking platform using the MERN stack (MongoDB, Express.js, React, Node.js), supporting thousands of concurrent users and delivering accurate live telemetry and fleet performance insights.
- Designed and deployed cloud-native infrastructure on AWS, leveraging S3, Lambda, API Gateway, EC2, and Cognito, ensuring secure authentication, session scalability, and highly available backend services.
- Engineered real-time data streaming systems using WebSockets and Socket.io, enabling instant map updates, route tracking, and telemetry broadcasting with low-latency data handling.
- Developed secure authentication and authorization layers using OAuth2, JWT, and Multi-Factor Authentication (MFA), enhancing system resilience and data protection.
- Built scalable, reusable UI components with React + TypeScript, and implemented advanced state management patterns with Redux and Context API to ensure consistent and performant frontend behavior.
- Designed and implemented RESTful APIs and GraphQL endpoints for modular data consumption, aligning backend services with scalable microservice communication models.
- Optimized MongoDB performance using compound indexes, sharding, schema validation, and Mongoose ORM, improving data retrieval speeds and reducing query load under high traffic.
- Implemented CI/CD pipelines using Jenkins, GitLab CI, and Docker, ensuring automated testing, build, and zero-downtime deployment across staging and production environments.
- Enabled infrastructure-as-code (IaC) practices using Terraform and AWS CloudFormation, standardizing environment provisioning and improving deployment consistency.
- Led development of interactive dashboards using custom charting libraries and Map APIs, providing live visual analytics for vehicle diagnostics, trip history, and route efficiency.
- Applied performance tuning techniques such as code splitting, lazy loading, and caching (Redis), significantly reducing initial load times and improving real-time responsiveness.
- Authored and maintained comprehensive technical documentation, API references, architectural diagrams, and operational guides for development and DevOps teams.
- Championed Agile and Scrum methodologies, conducting sprint planning, retrospectives, and code reviews while mentoring junior engineers in best practices, scalable architecture, and clean code principles.
- Conducted extensive automated and manual testing using Jest, Cypress, Mocha, and Postman, ensuring robust QA coverage across backend and frontend services.

## **Zillow - Florida - Full stack engineer Jul 2018 - Sep 2019**

- Designed and developed enterprise-level real estate platforms using the MERN stack (MongoDB, Express.js, React, Node.js), delivering intuitive property search, listings, and management systems.
- Architected and deployed cloud-native infrastructure on AWS (S3, Lambda, API Gateway, EC2), ensuring fault-tolerant, scalable backend services and efficient serverless data workflows.
- Developed advanced frontend features such as interactive map visualizations, dynamic filters, and real-time property listings, enhancing UX across web and mobile interfaces.
- Built and secured RESTful APIs for real-time data access and third-party integrations, including payment gateways, geolocation services, and MLS property data providers.
- Implemented SSR (Server-Side Rendering) with React and Next.js, lazy loading, and intelligent caching strategies to significantly improve SEO and application performance.
- Optimized complex client-side state using Redux and Context API, enabling responsive UI behavior and consistent user session experiences.
- Integrated TypeScript and enforced modern development standards, improving maintainability, scalability, and type safety across full stack systems.
- Collaborated with product owners and UI/UX teams under Agile and Scrum methodologies to prioritize backlogs, estimate workloads, and meet sprint deliverables.
- Led code reviews and provided technical mentorship on front-end best practices, clean architecture, and efficient component design patterns.
- Authored detailed system documentation, architectural flow diagrams, and deployment guides to support onboarding, scalability, and cross-team alignment.
- Automated testing workflows using Jest, Cypress, and Postman, ensuring high code coverage, regression protection, and reliable production releases.
- Initiated performance audits and implemented database indexing, query optimization, and CDN integration, reducing latency and load times across key workflows.